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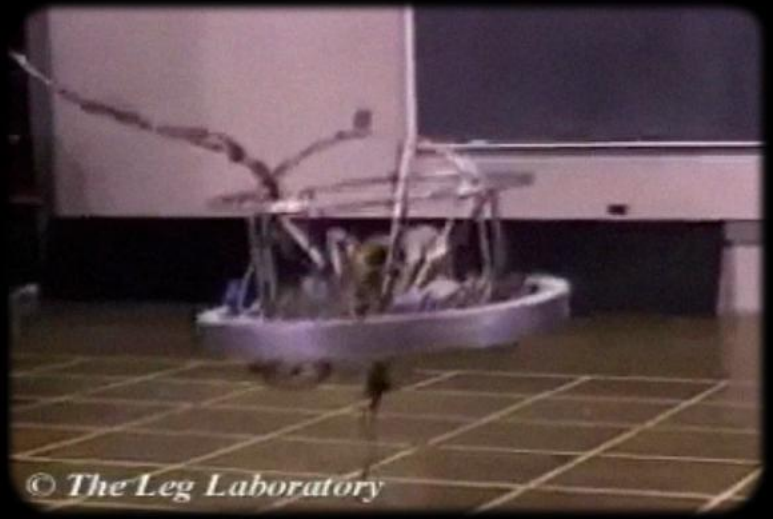
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Everything AI: The Path to Sustainability

Robert Playter

Former CEO, Boston Dynamics

Ministry of Education THE KOREA ECONOMIC DAILY KRIVET Korea Research Institute for Vocational Education & Training


© The Leg Laboratory



AI is Revolutionizing Robotics

Personal History



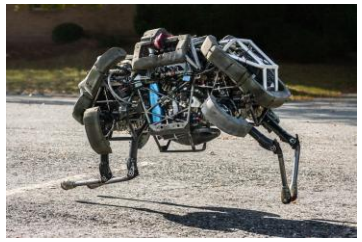
MIT Leg Lab in 1992



Robert in 1985

A History Of Robotics Innovation

Company Founded

1992	2014	2018	2021
 *LittleDog*	 *BigDog*	 *LS3*	 *Cheetah*	 *Spot*	 *Spot V2*	 *Spot*	
 *Petman Prototype*	 *Petman*	 *Atlas*	 *Next Gen Atlas*	 *Next Gen Atlas*	 *Electric Atlas*
 *Rhex*	 *SandFlea*	 *Handle*	 *Handle*	 *Stretch*


BostonDynamics
LEVATAS

Mobility Has Been Solved



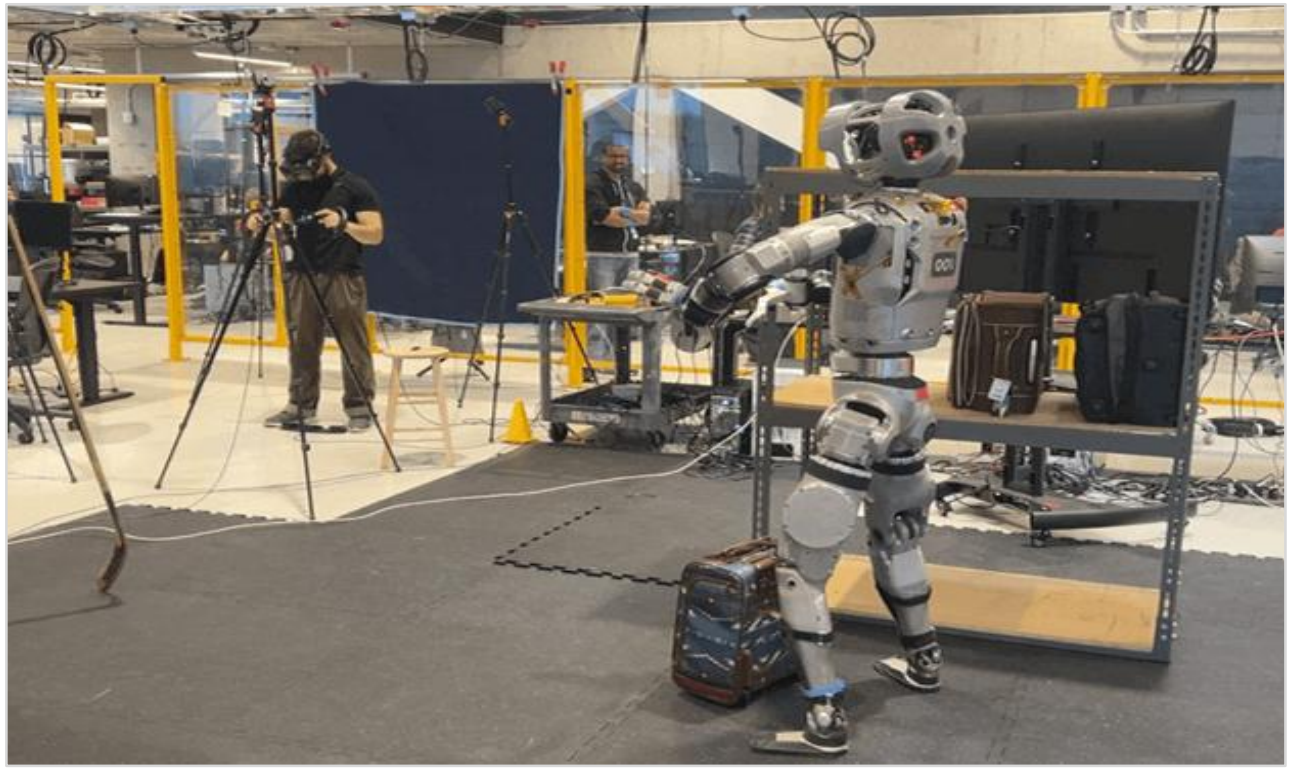


ANYmal RESEARCH

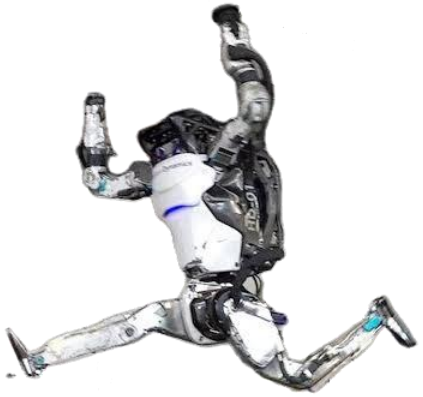
UNITREE

AI Enables Physically Intuitive Training

- Skilled workers can directly train robots
- No programming required
- Detailed knowledge of skills is required
  - Successes
  - Failures
- Knowledge of robot capabilities & limitations is required



Progression of Humanoid Robot Tasks



Bodily Motion | **Pick & Place** | **Assembly** | **Troubleshooting**

Increasing Dexterity & Physical Diagnosis

Easy → **Hard**

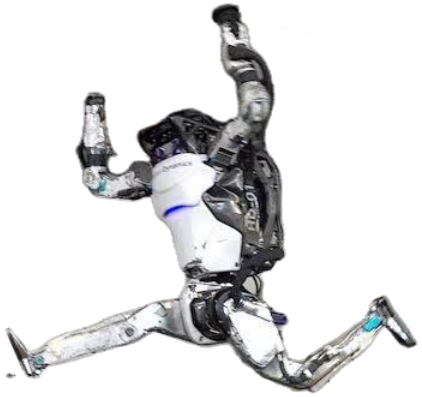
Hyundai Will Deploy Atlas at Scale

HYUNDAI MOTOR GROUP

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Progression of Humanoid Robot Tasks



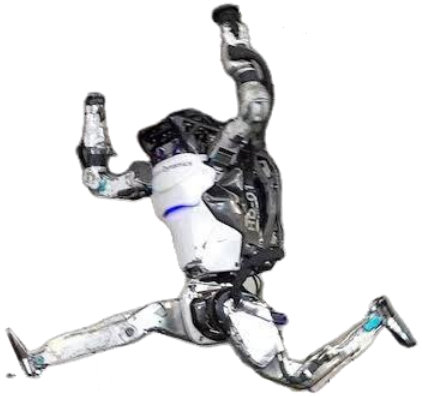


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Progression of Humanoid Robot Tasks





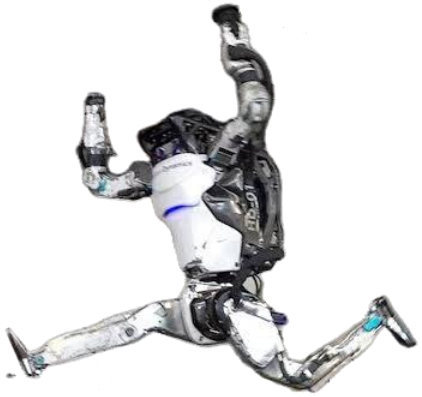


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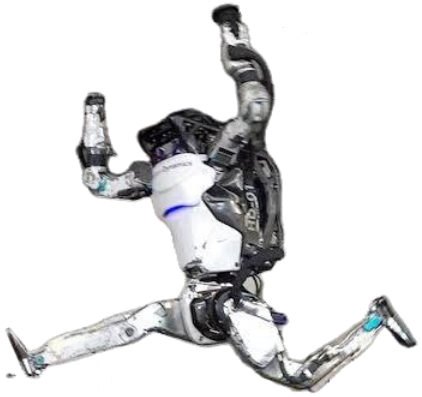


Bodily Motion | **Pick & Place** | **Assembly** | **Troubleshooting**

Increasing Dexterity & Physical Diagnosis

Easy — **Hard**

Progression of Humanoid Robot Tasks









Bodily Motion | **Pick & Place** | **Assembly** | **Troubleshooting**

Increasing Dexterity & Physical Diagnosis

Easy → **Hard**

Dexterous Manipulation Remains a Challenge

- Large scale data for dexterous manipulation AI training does not exist
  - Unlike speech and image recognition
- Human manipulation skills depend upon sensing that is orders of magnitude more complex than is currently possible in hardware
- Physics-based simulation is not yet capable of capturing the details of dexterous manipulation
- It is likely 10 years before this is solved



Market Progression







ROBOT PRICE

TASK COMPLEXITY

Robots Will Enhance the Welfare of Workers

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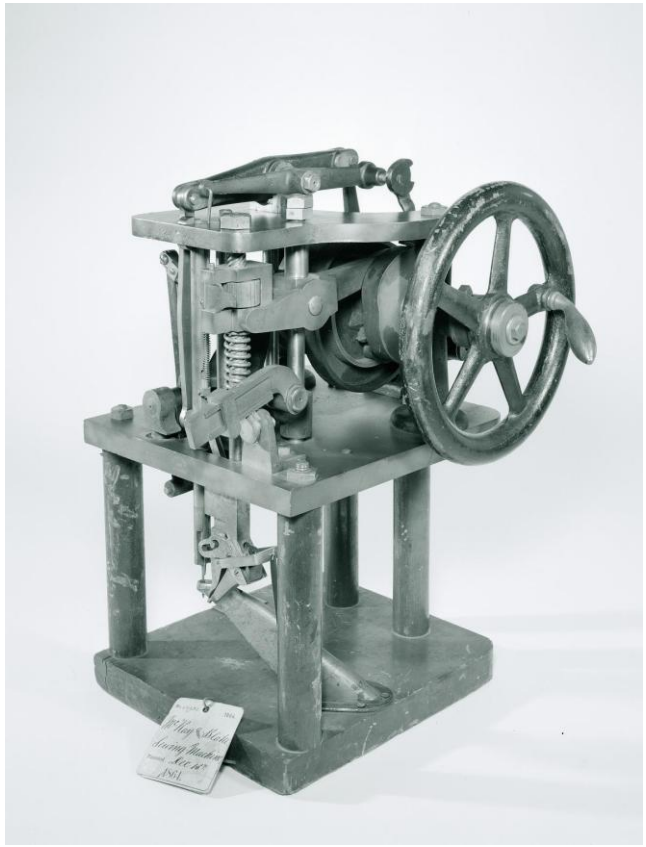






Historical Precedents for Technological Innovation







Spinning Jenny, 1770's	
Cost	15% of annual wages
Productivity Increase	3X
Wage Increase	2X

Shoemaking Machine, 1860's	
Cost	15-25% of annual wages
Productivity Increase	6X
Wage Increase	2X

Chainsaw Logging, 1950's	
Cost	30% of annual wages
Productivity Increase	2X
Wage Increase	2X

How to Ensure Workers Can Benefit

- Train people how to work with robots
  1. What is easy and hard for robots (or people)
  2. How to train physical AI: successes & common failures
  3. Standardize interactions with different kinds of robots
- Make the technology easy to own & operate
  1. Portable
  2. Inexpensive (25% of annual wages)
  3. Generalizable to new tasks
- Localized & personalized services
  1. High labor content
  2. Small crews
  3. Local service market
  4. Skilled human + unskilled assistant
  5. Customer relationship belongs to worker

Summary

- Historically, automation has been shown to improve the quality of life for people. I believe mobile robots will also be good for humanity.
- The fear about robots is that the changes will come so quickly that people do not have time to adjust.
- Physical AI has accelerated robot development considerably, but hard problems remain that limit the pace they can reproduce all human labor. Physical AI will progress more slowly than pure software AI due to limitations in data sources and hardware challenges
- In the next 3 years you will begin to see scaled humanoid robot deployments in manufacturing and logistics operations
- In 10 years robots will be capable of much of our manual labor and will transform manufacturing work
- In 15-20 years, widely available and affordable robots will unleash an entrepreneurial explosion of small operators who buy robots to improve productivity in labor intensive businesses

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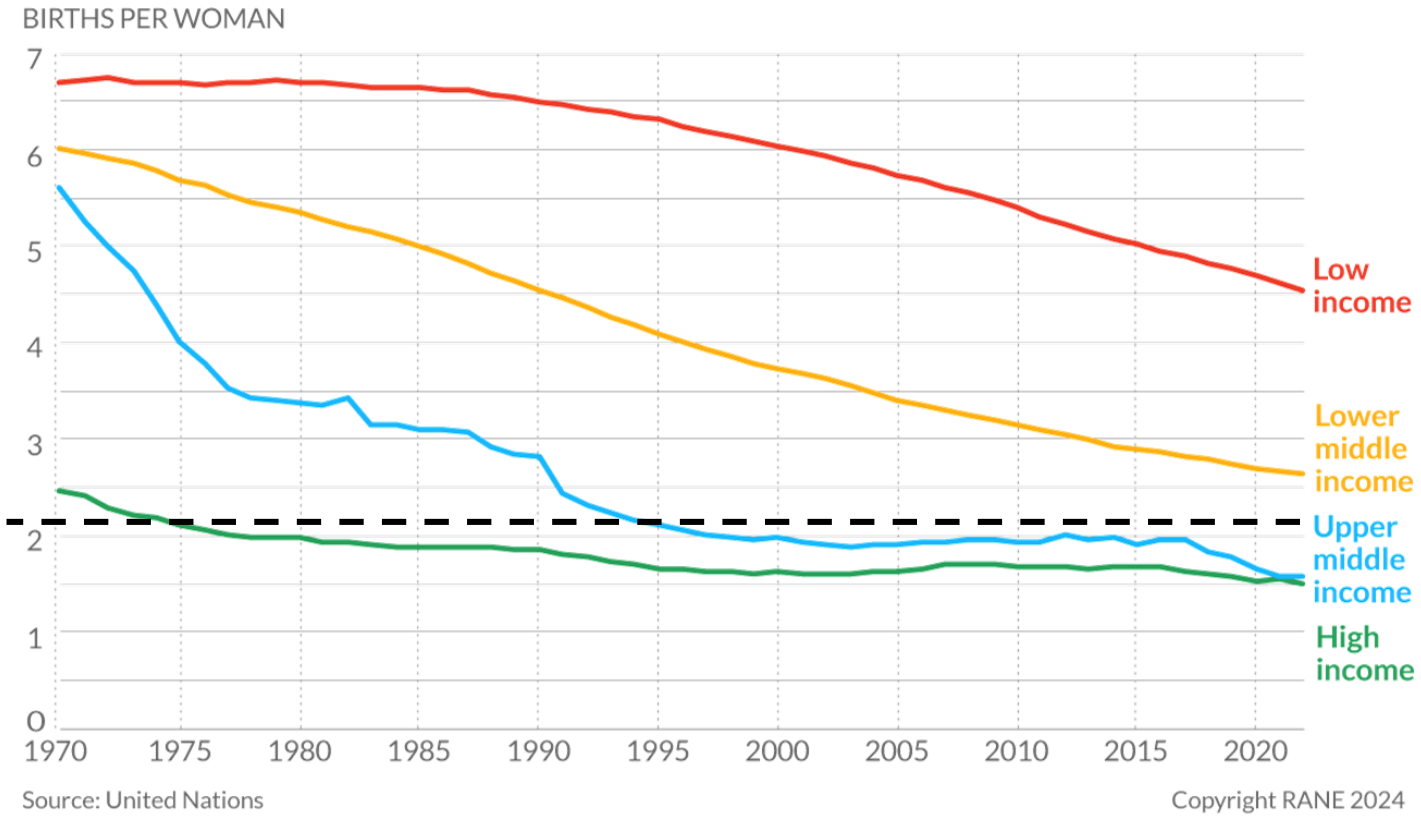
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Questions?

Upper Income Countries Are Losing Workers

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Fertility Rate by Country Income Group



Total global labor deficit in 2030[1]

$8.5T

[1]Unrealized output due to labor deficit $8.5 trillion; Source: Future of work report